

Loss Function Comparison

M1 Architecture

Loss Function	EER ↓	Rank-1 ↑	Verdict
SupCon	0.89%	76.23%	★ Best verification
AdaCos	3.15%	76.68%	Best Rank-1, much worse EER
ArcFace	3.14%	63.68%	Poor
SubCenter ArcFace	3.54%	69.51%	Worst EER

M2 Architecture

Loss Function	EER ↓	Rank-1 ↑	Verdict
SupCon	0.90%	79.82%	★ Best verification
AdaCos	1.79%	90.13%	★ Best identification
ArcFace	3.08%	71.75%	Poor
SubCenter ArcFace	4.01%	63.68%	Worst

G2 Architecture

Loss Function	EER ↓	Rank-1 ↑	Verdict
SupCon	0.94%	80.72%	★ Best overall
ArcFace	3.53%	76.23%	Moderate
SubCenter ArcFace	3.66%	66.82%	Poor
AdaCos	4.93%	78.92%	Worst verification

Overall Comparison

Architecture	Best Verification Loss	EER	Best Identification Loss	Rank-1
M1	SupCon	0.89%	AdaCos	76.68%
M2	SupCon	0.90%	AdaCos	90.13%
G2	SupCon	0.94%	SupCon	80.72%

Verification Ranking (Lowest EER)

Rank	Model	Loss	EER	Rank-1
	M1A	SupCon	0.89%	76.23%
	M2A	SupCon	0.90%	79.82%
	G2A	SupCon	0.94%	80.72%
4	M2B	AdaCos	1.79%	90.13%
5	M2B	ArcFace	3.08%	71.75%
6	M1B	ArcFace	3.14%	63.68%
7	M1B	AdaCos	3.15%	76.68%
8	G2B	ArcFace	3.53%	76.23%
9	M1B	SubCenter	3.54%	69.51%
10	G2B	SubCenter	3.66%	66.82%
11	M2B	SubCenter	4.01%	63.68%
12	G2B	AdaCos	4.93%	78.92%

Identification Ranking (Highest Rank-1)

Rank	Model	Loss	Rank-1	EER
	M2B	AdaCos	90.13%	1.79%
	G2A	SupCon	80.72%	0.94%
	M2A	SupCon	79.82%	0.90%

Rank	Model	Loss	Rank-1	EER
4	G2B	AdaCos	78.92%	4.93%
5	M1B	AdaCos	76.68%	3.15%
6	M1A	SupCon	76.23%	0.89%
7	G2B	ArcFace	76.23%	3.53%
8	M2B	ArcFace	71.75%	3.08%
9	M1B	SubCenter	69.51%	3.54%
10	G2B	SubCenter	66.82%	3.66%
11	M1B	ArcFace	63.68%	3.14%
12	M2B	SubCenter	63.68%	4.01%

Key Findings

Observation	Conclusion
Best verification loss	SupCon
Best identification loss	AdaCos (only for M2)
ArcFace performance	Consistently inferior to SupCon
SubCenter performance	Worst overall across architectures
Lowest EER	M1A + SupCon (0.89%)
Highest Rank-1	M2B + AdaCos (90.13%)